

Stat 150 – Stochastic Processes

Comprehensive exercise solutions

Durrett, *Essentials of Stochastic Processes* (3rd ed.) ■ Fall 2026 ■ Keshav Ramamurthy

Adding a solution

To record a solved exercise, type under its chapter banner:

```
\exercise{1.4.2} \begin{solution} ... \end{solution}
```

Restate the problem first with `\begin{problem}[title]... \end{problem}`; structure proofs with `claim`, `idea`, `proofsketch`.

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Part I Durrett, *Essentials of Stochastic Processes*

Chapter 1 Markov Chains

Durrett, §1.1–1.13: definitions, multistep transitions, classification of states, stationary distributions, limit behavior, exit distributions and times

Chapter 2 Poisson Processes

Durrett, §2.1–2.6: exponential distribution, constructing the process, compound Poisson, thinning/superposition/conditioning

Chapter 3 Renewal Processes

Durrett, §3.1–3.5: laws of large numbers, queueing applications, age and residual life

Chapter 4 Continuous Time Markov Chains

Durrett, §4.1–4.8: generators, transition probabilities, limiting behavior, exit times, Markovian queues

Chapter 5 Martingales

Durrett, §5.1–5.5: conditional expectation, gambling strategies and stopping times, applications

Chapter 6 Mathematical Finance

Durrett, §6.1–6.3: binomial model, Black–Scholes, options

Part II Pinsky & Karlin, *An Introduction to Stochastic Modeling*

Chapter 1 Introduction

Pinsky & Karlin: stochastic modeling, probability review

Chapter 2 Conditional Expectation and Related Topics

Pinsky & Karlin: conditioning, generating functions, martingales

Chapter 3 Markov Chains

Pinsky & Karlin: discrete-time chains, first-step analysis

Chapter 4 The Long Run Behavior of Markov Chains

Pinsky & Karlin: stationary and limiting distributions, absorption

Chapter 5 Poisson Processes

Pinsky & Karlin: the Poisson process and its transformations

Chapter 6 Continuous-Time Markov Chains

Pinsky & Karlin: birth–death processes, generators, branching processes

Chapter 7 Renewal Processes

Pinsky & Karlin: renewal theory and applications

Chapter 8 Brownian Motion and Gaussian Processes

Pinsky & Karlin: Brownian motion, drift, variants

Chapter 9 Queueing Systems

Pinsky & Karlin: Markovian queues, Little's formula, networks